

1. Behavior cloning

Training sample: (o_t, ℓ_i, a_t)

$$\mathcal{L}_{\text{BC}} = -\log p_{\theta}(a_t \mid o_t, \ell_i)$$

learns expert action likelihood under the given task instruction.

2. Counterfactual instruction pair

Keep the same observation and action: (o_t, a_t)

correct instruction ℓ_i
score $s^+ = \log p_{\theta}(a_t \mid o_t, \ell_i)$

wrong-target instruction ℓ_i^-
score $s^- = \log p_{\theta}(a_t \mid o_t, \ell_i^-)$

The negative changes the task target, not the image or expert action.

3. Training objective

$$\mathcal{L} = \mathbb{E}[w_i \mathcal{L}_{\text{BC}}] + \lambda[m - (s^+ - s^-)]_+.$$

Rare tasks use a capped positive weight w_i .

TCAD is applied only to eligible sampled pairs.

Inference: unchanged VLA policy, no generator, detector, or extra module.